Zhenyi Zhang

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PhD Candidate in Space Geodesy, ETH Zurich

ETH Profile | Google Scholar

Education

ETH Zurich — Zurich, Switzerland

PhD in Space Geodesy — *10.2023 – 10.2027 (expected)*[1]

Institute of Geodesy and Photogrammetry Supervisor: Prof. Dr. Benedikt Soja

Thesis: Physics-Informed Machine Learning for Tropospheric Modeling

Wuhan University — Wuhan, China

MSc in Geodesy and Surveying Engineering, Distinguished Graduate — *09.2020 – 06.2023*

Thesis: Processing and Applications of High-Precision Tropospheric Products from Airborne GNSS

Wuhan University — Wuhan, China

BSc in Navigation Engineering, Distinguished Graduate — *09.2016 – 06.2020*

GPA: 3.98/4.0, rank 1/51

Research Profile

My research lies at the intersection of **physics-informed machine learning, space geodesy, and atmospheric science**. My doctoral research focuses on developing physically constrained AI models for tropospheric modeling by integrating geodetic observations, meteorological data, and physical knowledge. My broader expertise includes precise GNSS processing for ground-based and high-dynamic airborne platforms, tropospheric delay estimation and modeling, GNSS data assimilation in numerical weather prediction, and atmospheric correction for remote sensing. A major part of my work focuses on developing **global physics-constrained machine learning models** for tropospheric mapping functions, slant wet delays, and zenith wet delays by combining geodetic formulations, ERA5 atmospheric fields, and numerical ray tracing. I also contribute to the Earth System Foundation Model project, exploring the integration of GNSS-calibrated satellite observations, meteorological analyses, and other Earth-system data into AI foundation models for atmospheric water-vapor estimation and weather-related applications.

Selected Publications

Full publication list: Google Scholar

- **Z. Zhang** and B. Soja (2026). Machine-learned mapping functions for the asymmetric troposphere. ***Journal of Geodesy***, 100, 79. doi:10.1007/s00190-026-02101-5.
- **Z. Zhang** and B. Soja (2026). MSWD: A hybrid machine-learning framework for slant wet delay modeling. ***Journal of Geophysical Research: Atmospheres***. doi:10.1029/2026JD046410.
- **Z. Zhang**, M. Liu, W. Zhang, Y. Lou, and B. Soja (2025). Assimilating ground-based and high-dynamic airborne GNSS zenith total delays into numerical weather predictions. ***IEEE Transactions on Geoscience and Remote Sensing***. doi:10.1109/TGRS.2025.3583462.
- **Z. Zhang**, Y. Lou, W. Zhang, Z. Wang, Y. Zhou, J. Bai, Z. Zhang, and C. Shi (2023). Dynamic stochastic model for estimating GNSS tropospheric delays from airborne platforms. ***GPS Solutions***. doi:10.1007/s10291-022-01375-4.
- **Z. Zhang**, Y. Lou, W. Zhang, H. Wang, Y. Zhou, and J. Bai (2022). Assessment of ERA-Interim and ERA5 reanalysis data on atmospheric corrections for InSAR. ***International Journal of Applied Earth Observation and Geoinformation***. doi:10.1016/j.jag.2022.102822.
- **Z. Zhang**, Y. Lou, W. Zhang, Y. Ouyang, Y. Zhou, J. Bai and B. Soja (2026), Assessing the Impact of Dynamic Tropospheric Stochastic Constraints on Airborne GNSS Positioning and ZTD Estimation. ***Measurement*** doi:10.1016/j.measurement.2026.123448.
- **Z. Zhang**, Y. Lou, W. Zhang, H. Wang, Y. Zhou, J. Bai (2021), On the Assessment GPS-Based WRFDA for InSAR Atmospheric Correction: A Case Study in Pearl River Delta Region of China. ***Remote Sensing***.

[1] Expected completion of PhD defense: July 2027; formal degree expected in October 2027.

doi:10.3390/rs13163280

- **Z. Zhang**, Y. Cheng, F. Ozdemir, L. Trentini and B. Soja (2026), PhyVapor-ESFM: A Physics-Constrained Framework for Integrating Satellite Water Vapor Observations into Foundation Model. ***Workshop on Machine Learning for Earth System Modelling*** (24–26 August 2026 · Bonn, Germany).
- F. Ozdemir, Y. Cheng, S. Mohebi, F. Lehmann, S. Adamov, **Z. Zhang**, L. Trentini, D. Grund, O. Fuhrer, T. Hoefler, S. Mishra, S. Schemm, B. Soja and M. Salzmann (2026), Earth System Foundation Model (ESFM): A unified framework for heterogeneous data integration and forecasting. ***Communications Earth & Environment (Revising)***. arXiv preprint :2605.00850

Research and Academic Activities

- Contributor to the **Swiss AI Initiative** and the **Earth System Foundation Model (ESFM)** project, working on atmospheric water vapor and the integration of geodetic, satellite, and meteorological observations.
- Supervised five bachelor and master theses/projects on GNSS meteorology, machine learning, atmospheric water vapor, and climate-related applications.
- Teaching Assistant at ETH Zurich for graduate courses *Geodetic Earth Monitoring* and *Geospatial Reference Systems*.
- Reviewer for *Journal of Geodesy*, *Journal of Advances in Modeling Earth Systems*, *IEEE Transactions on Geoscience and Remote Sensing*, *Atmospheric Research*, *Earth and Space Science*, and other journals.

Selected Honors and Awards

- National Scholarship for Graduate Students, China (top 0.2% of graduate students nationwide)
- National Scholarship for Undergraduates, China (top 0.2% of undergraduate students nationwide)
- Wang Zhizhuo Innovative Talent Scholarship, Wuhan University (8 recipients annually in surveying, geodesy, and remote sensing)
- Top Ten Youths Honor, Wuhan University (10 recipients annually university-wide)

Supervised Student Theses and Projects

- Leonard Haas (2026, Master thesis): Assessing GNSS ZWD and Horizontal Gradients as Indicators of Global Climate Change.
- Jingru Tong (2026, Master thesis): Feasibility of Using Global Ground-Based GNSS PWV as an Early-Warning Signal for ENSO.
- Zhihe Yu (2025, Bachelor thesis): Feasibility Study of El Niño-Southern Oscillation Prediction Using GNSS Tropospheric Products and Machine Learning.
- Kaan Marti (2026, Master project): Towards more accurate ML-based fine-tuned slant wet delay models.
- Zhihao Cai (2026, Master project): Global Hybrid Machine-Learning Models for Slant Water Vapor.

Expertise and Other Skills

- **Expertise**: Space geodesy, GNSS data processing, Machine learning/Deep Learning, GNSS meteorology, Tropospheric modeling, InSAR data processing, radiosonde data processing, Numerical weather prediction and data assimilation, Foundation models for Earth system science
- **Languages**: Chinese (native speaker), English (fluent)
- **Programming**: Matlab (proficient), Python (proficient), C/C++ (proficient), Fortran (proficient)